Looking for a frequency-based principle to predict the sensitivity of convolutional neural networks to Gaussian image perturbations*

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Abstract. Distribution shifts, i.e. breaks of the i.i.d. assumption, unpredictably alter the performance of convolutional neural networks (CNNs). It limits their practical ability, as distribution shifts are widespread in real-life applications, and understandable robustness to these shifts is necessary, especially in safety-critical applications. We propose to leverage Fourier analysis to explain the sensitivity of CNNs to Gaussian blur and noise. These shifts models image acquisition conditions. Acquisition conditions play a pivotal role in the lack of reliability of CNN-based detectors deployed for solar array detection. Simple experiments show that the same Gaussian perturbation has an uneven impact on the input. We propose to investigate whether the frequency principle can help us understand why CNNs are less robust to Gaussian perturbation for some inputs rather than others. Following the frequency principle, we expect to see greater robustness when images are learned at early stages of training. Results only partially align with the prediction. It motivates further research to identify how a model correlates image frequencies and image labels to understand the robustness to Gaussian perturbation.

Keywords: Distribution shifts · Robustness · Convolutional neural networks · Frequency principle · spectral bias

1 Problem statement

Despite remarkable performance, convolutional neural networks (CNNs) experience a sharp performance drop when exposed to distribution shifts, i.e., when the i.i.d. assumption is broken between the training and testing datasets[10]. Distribution shifts are ubiquitous in real-world applications [18]. In the case of photovoltaic (PV) installations mapping, [8] showed that the performance drop

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could be as large as 30 percentage points when shifting from the training dataset to the *mapping area*, i.e., the data on which the mapping pipeline is deployed in production.

Lack of robustness regarding distribution shifts causes this performance drop[19]. We define robustness as the ability of an algorithm to achieve similar performance if the train and test samples are close [17]. In our case, we hypothesize that the lack of robustness towards varying acquisition conditions across the *mapping area* explains the performance drop. Acquisition conditions encompass optical system heterogeneity, brightness, noise, and blurriness. Image corruptions such as those proposed by [6] conveniently model acquisition conditions. We focus on Gaussian blur as it models the optical system.

While Gaussian blur is known to degrade the performance of CNNs [5], simple experiments show that the perturbation does not affect *all* samples equally: after blurring, some samples remain accurately predicted while others are no longer recognized. Since Gaussian blurring affects the intensity frequencies contained in the input image, we aim to propose a framework explaining why the effects of Gaussian perturbations are uneven across data samples based on Fourier analysis.

2 Why this problem is important

Acquisition conditions in remote sensing In applications like ours - solar panel detection - acquisition conditions play a crucial role since solar panels often appear as small objects in complex scenes [11]. Therefore, the sensitivity of CNN-based detectors to image quality is crucial compared to other cases where objects are large. We suspect that the lack of robustness towards acquisition conditions is essential in explaining why CNN models trained for PV panel classification experience a sharp performance drop over the *mapping area*.

Deep neural networks in safety-critical applications Moreover, understanding the effect of distribution shifts on deep neural networks (DNN) is critical to foster safer real-life deployment [16]. Recent works have shown how DNNs can be sensitive to different ethnicities [2] or lightning conditions [3]. For remote PV detection, [4] identifies the lack of reliability as the principal limit to using CNN models in producing official statistics. We wish to provide a framework explaining *why* CNNs lack robustness regarding acquisition conditions.

3 Why existing solutions in the literature are insufficient

3.1 Related literature

Robustness of CNNs in remote sensing While many works propose methods to mitigate the effects of distribution shift in remote sensing (see [14] for a review), very few work-study how shifts affect performance. [15] focused on the

effect of the geographical shift and [11] on the effects of different image characteristics (non-concentrated distribution, ground sampling distance) on accuracy. However, these studies only provide practical takeaways and no possible mechanisms explaining why the identified factors impacted accuracy. Such takeaways are not sufficient when it comes to developing more trustworthy systems.

Robustness The literature on image corruptions is well-established, with many studies and benchmarks [6, 10], as well as mitigation strategies [19, 7]. However, recent research showed that mitigation strategies could have adverse effects: [1] showed that data augmentation increased accuracy at the expense of certain classes in the dataset. Therefore, it is necessary to understand what causes the lack of robustness rather than proposing new mitigation strategies.

3.2 Our contribution

Our main contribution is to explain why Gaussian perturbations affect CNNs. It contributes to the literature on robustness as it better explains why CNNs are not robust to a given class of distribution shifts. It also contributes to the literature in remote sensing as it paves the way for more robust PV mapping models.

4 Our approach

4.1 Setting

We set ourselves in a binary classification task. We use the dataset of distributed PV installations introduced by [9]. This dataset is nearly balanced (around 45% of positive labels and 55% of negative labels). We want to identify solar panels on overhead imagery. We consider a ResNet-50 model trained via ERM. We evaluate the model's accuracy on a held-out test set, which comes from the same distribution as the training and validation sets.

We test robustness to corruption by altering the test set, evaluating the model on the altered test set, and computing the consistency metric introduced by [19]. The consistency measures how often the network outputs the same classification, given the same image with two different shifts. To apply a Gaussian blur, we consider an image I from the test dataset and modify it as follows to generate an altered image I': $I' = g_\sigma * I$ where g is a Gaussian filter parametrized by σ. σ varies between 0.1 and up to 20 in our experiment with the frequency principle detailed in section 5.1

4.2 Gaussian perturbations and pivotal frequencies

As a baseline experiment, we apply a random Gaussian perturbation to the test set and evaluate the model on this altered test set using consistency. Our overall consistency between the unaltered and altered images of the test dataset

is 98%. This means that all images have not been affected the same way by the transformation. The σ applied to the Gaussian perturbation is randomly picked between 0.1 and 1. For a human viewer, it is still possible to see the array with the largest σ.

We study how the output probabilities shift for images that the model mispredicts based on this result. We witness that for these images, the output probability shifts: the model predicts no PV panel with high probability, whereas prior to the alteration, it predicted a PV panel with high probability.

We interpret this result as follows: (i) Gaussian perturbations affect the frequency components in the image. (ii) the model should correlate frequencies in the image with the label, and depending on the images, the frequencies must be different (iii) when the prediction is *flipped*, it means that the transformation has removed the frequency which the model correlates with the label. We call this frequency the *pivotal* frequency.

5 Some preliminary experiments

5.1 Robustness to Gaussian perturbations and frequency principle

We first tested whether the *frequency principle*[13] can explain the inhomogeneous robustness to Gaussian blurring. The spectral bias or frequency principle explains the training dynamics of neural networks. According to this principle, deep learning models fit the frequencies of the function linking the inputs and the outputs from the lowest to the highest during training.

Working hypothesis and framework According to this principle, the *sooner* the model learns an image, the more robust the prediction to a Gaussian blurring. Indeed, if the model learns an image at an early training stage, then it should be based on lower frequencies. We need to record when images are "learned." We consider that an image of the test set is learned once it gets correctly predicted by the model. We train the model from scratch and evaluate it regularly on the test set to highlight when images are learned. We list the test set's correctly predicted images at each evaluation (labeled by the gradient iteration t).

Once training is complete, we consider the trained model. We apply a gradually stronger Gaussian blur on the test images and record what parameter σ is necessary to "flip" the prediction. We expect a negative correlation between the moment t an image was learned and the strength of σ necessary to flip the prediction. In other words: *images learned later will tend to be less robust to Gaussian blur and noise than images learned earlier.*

Results partially align with the theory Results are displayed on figure 1. As expected, there is a negative correlation between robustness to perturbations and the time necessary to predict an image correctly. *But*, we found that the robustness-earliness correlation only holds on average. Many images that are learned early are as fragile as images learned later. We observe that the range of strength gradually concentrates towards a given value.

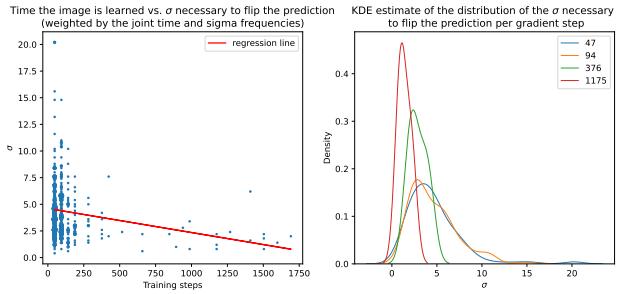

Fig. 1. Left: the relationship between the moment an image of the test set is correctly predicted (x-axis) and the strength of the variance parameter σ necessary to fool the model (y-axis). σ varies between 0 and 20. Blue dots correspond to images whose size is proportional to the number of images. The red line corresponds to the regression line. Right: Distribution (KDE estimate) of the σ's that successfully flipped the model's prediction. Each curve represents the distribution of σ's at one time step t.

5.2 Take-aways and future works

Our experiment based on the frequency principle theory confirmed that altering the frequency components of an image alters the prediction. It comforts us with the idea that we can derive a Fourier-based explanation of Gaussian robustness. However, our experiment revealed that the training dynamics are not very helpful in predicting which frequency components are fitted to the label.

In future work, we plan to:

- Determine whether the model can change its predictive frequencies during training. Candidates include the gradient starvation [12] ;
- Quantify the frequencies the model learns and discuss their interplay with the training and altered datasets ;
- To further assess the generality of our results by considering other corruptions. Our results with the frequency principle also hold with Gaussian noise. We will also consider alternative datasets, such as [6].

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